Do Now · retrieval

Lesson 090 · Inside City Hall: ALU, Control Unit and Registers

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does CU stand for?

Q2

What is the ALU?

Q3

What does ACC typically store?

Q4

A full adder has A=1,B=1 and carry-in=0. What are Sum and carry-out?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — Control Unit generates control signals and interprets the fetched instruction. It is not the component that performs arithmetic itself.

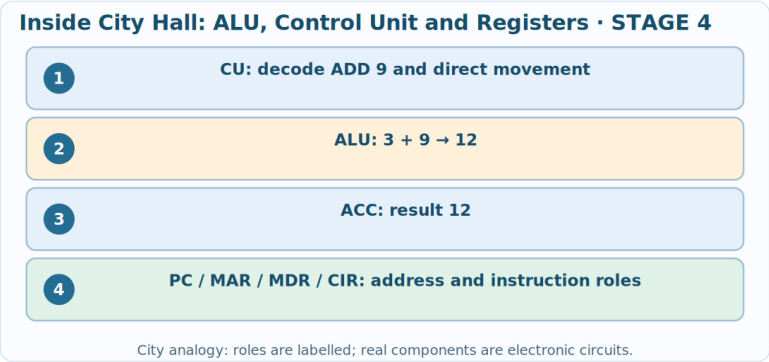
Core idea 2

Explain — ALU completes arithmetic and logic such as addition and comparisons; the ACC may hold a result in an accumulator-style model.

Core idea 3

Evaluate — Registers serve distinct roles: PC next instruction, MAR memory address, MDR transferred contents, CIR current instruction, ACC intermediate calculation.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

Control Unit — Directs instruction execution.

ALU — Performs arithmetic and logic.

ACC — Intermediate calculation register.

PC — Address of next instruction.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

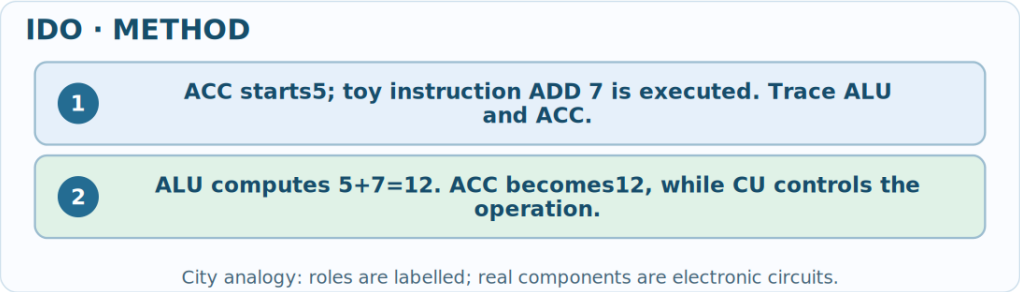
Problem

ACC starts5; toy instruction ADD 7 is executed. Trace ALU and ACC.

Follow the worked method

1. Model response — ACC holds intermediate result, not memory address.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Model response
Do not say PC is overwritten by subtraction; it tracks instruction address.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

For ACC=3 and toy ADD9, state ALU calculation, ACC result, CU role. [6]

Question 2

Explain different roles of PC, MAR and CIR in a memory fetch. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Pupil claims CU performs calculations while ALU stores next instruction address. Correct both errors. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

The Program Counter is the ALU register storing calculation results.

Use key vocabulary from memory

1. Explain Control Unit in your own words.

2. Explain ALU in your own words.

3. Explain ACC in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.